1. Define Artificial Intelligence and list its major areas.

Artificial Intelligence (AI) is a branch of computer science that enables machines to perform tasks that normally require human intelligence, such as learning, reasoning, and decision-making.

Major Areas of AI:

1. Machine Learning
2. Natural Language Processing (NLP)
3. Robotics
4. Computer Vision
5. Expert Systems
6. Knowledge Representation

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2. Discuss the history and evolution of AI.

- **1950** – Alan Turing introduced the *Turing Test*.
- **1956** – AI term introduced at Dartmouth Conference.
- **1960s–70s** – Development of expert systems.
- **1980s** – AI winter (less funding due to slow progress).
- **1990s** – Growth of machine learning.
- **2000s–Present** – Deep learning, big data, AI in healthcare, finance, robotics, etc.

AI evolved from rule-based systems to modern data-driven deep learning models.

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3. Explain Intelligent Agents and their architecture.

An **Intelligent Agent** is an entity that perceives its environment through sensors and acts upon it using actuators.

Types of Agents:

1. Simple Reflex Agent
2. Model-Based Agent
3. Goal-Based Agent
4. Utility-Based Agent

5. Learning Agent

Architecture Components:

- Sensors (Input)
- Environment
- Agent Program
- Actuators (Output)

Structure:
Sensors → Agent → Actuators

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4. What are AI applications in E-Commerce and Medicine?

In E-Commerce:

- Product recommendation systems
- Chatbots (customer support)
- Fraud detection
- Demand forecasting
- Personalized advertising

In Medicine:

- Disease diagnosis
- Medical image analysis
- Drug discovery
- Robotic surgery
- Patient monitoring systems

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5. Discuss AI techniques and knowledge representation methods.

AI Techniques:

- Search algorithms (BFS, DFS, A*)
- Machine Learning
- Neural Networks

- Fuzzy Logic
- Genetic Algorithms

Knowledge Representation Methods:

- Semantic Networks
- Frames
- Production Rules
- First-Order Logic
- Decision Trees

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6. Describe the issues in design of search algorithms.

Important issues:

1. **Completeness** – Does it always find a solution?
2. **Optimality** – Does it find the best solution?
3. **Time Complexity** – How fast does it work?
4. **Space Complexity** – How much memory is required?
5. **Branching Factor** – Number of child nodes per state.
6. **Heuristic Function Quality** – Accuracy of estimation.

Unit 2

7. Explain Generate-and-Test and Hill Climbing methods.

Generate-and-Test Method

- It generates possible solutions one by one.
- Each generated solution is tested to check whether it satisfies the goal condition.
- If solution is correct → stop.
- If not → generate another solution.

Steps:

1. Generate a possible solution.
2. Test whether it satisfies the goal.
3. Repeat until solution is found.

Advantage: Simple method
Disadvantage: Time-consuming if many possibilities exist

Hill Climbing Method

- It is a heuristic search technique.
- It moves in the direction of increasing value (better solution).
- At each step, choose the best neighbor.

Algorithm:

1. Start with initial state.
2. Evaluate neighboring states.
3. Move to the best neighbor.
4. Repeat until goal reached.

Problems:

- Local maxima
- Plateau
- Ridge problem

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8. Describe Best-First Search and A* Algorithm.

Best-First Search

- Expands the node which appears best according to heuristic function **h(n)**.
- Uses priority queue.
- Chooses node with lowest heuristic value.

Evaluation Function:

f(n)=h(n)f(n) = h(n)f(n)=h(n)

A* Algorithm

- Combines actual cost and heuristic cost.
- More efficient and optimal (if heuristic is admissible).

Formula:

f(n)=g(n)+h(n)f(n) = g(n) + h(n)f(n)=g(n)+h(n)

Where:

- **g(n)** = Cost from start to current node
- **h(n)** = Estimated cost to goal

Advantage: Optimal and complete (with good heuristic)

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9. Discuss Constraint Satisfaction and Means-Ends Analysis.

Constraint Satisfaction Problem (CSP)

- Problem where variables must satisfy certain constraints.
- Goal: Find values for variables that satisfy all constraints.

Components:

1. Variables
2. Domain
3. Constraints

Example: Sudoku, Map coloring

Means-Ends Analysis

- Problem-solving strategy.
- Reduces difference between current state and goal state.
- Selects operators to eliminate differences step-by-step.

Steps:

1. Compare current state and goal state.
2. Identify difference.
3. Choose operator to reduce difference.

4. Repeat until goal achieved.

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10. Explain Heuristic Search with suitable examples.

Heuristic search uses extra knowledge (heuristic function) to guide search.

Heuristic Function (h(n)):

- Estimates cost from current state to goal.

Examples:

1. **A* Search**
2. **Greedy Best-First Search**
3. **Hill Climbing**

Example (8-Puzzle):

- Number of misplaced tiles
- Manhattan distance

Advantage: Faster than blind search
Disadvantage: May not guarantee optimal solution

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11. Differentiate between A* and AO* Algorithms.

A* Algorithm	AO* Algorithm
Works on OR graph	Works on AND-OR graph
Finds shortest path	Finds optimal solution graph
Uses f(n)=g(n)+h(n)	Uses heuristic cost for AND-OR nodes
Expands one node at a time	May expand multiple nodes simultaneously
Used in path finding	Used in problem decomposition

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12. Advantages and Limitations of Search Techniques

Advantages

1. Systematic problem solving

2. Can find optimal solution
3. Useful in AI applications
4. Works for structured problems

Limitations

1. High memory requirement
2. Time complexity may be large
3. May get stuck in local optima
4. Depends on quality of heuristic

Unit 3

13. Define Knowledge Representation and its Importance

Definition:

Knowledge Representation (KR) is a method used in Artificial Intelligence to represent information about the world in a form that a computer system can understand and process.

It allows machines to store, organize, and use knowledge to solve problems.

Importance of Knowledge Representation:

1. Helps in decision making
2. Supports reasoning and inference
3. Improves problem-solving capability
4. Makes AI systems intelligent
5. Organizes complex information in structured form

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14. Explain Propositional Logic and Predicate Logic

Propositional Logic

- It deals with simple statements (propositions) that are either **True or False**.
- Uses logical operators like:
    - AND (∧)

    - OR (∨)
    - NOT (¬)
    - IMPLIES (→)

Example:
P: It is raining.
Q: Roads are wet.
P → Q

Limitation: Cannot express relationships between objects.

Predicate Logic (First-Order Logic)

- More powerful than propositional logic.
- Deals with objects, properties, and relationships.
- Uses:
    - Predicates
    - Variables
    - Quantifiers (∀ for "for all", ∃ for "there exists")

Example:
∀x (Human(x) → Mortal(x))

Advantage: Can represent complex statements.

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15. Different Approaches to Knowledge Representation

1. **Logical Representation**
    - Propositional Logic
    - Predicate Logic
2. **Semantic Network**
    - Graph-based representation of knowledge
3. **Frames**
    - Structured representation using slots and values
4. **Production Rules**

    - IF–THEN rules
5. **Conceptual Dependency**
    - Represents actions and relationships in structured way

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16. Discuss ISA Hierarchy and Frame Notation

ISA Hierarchy

- ISA means "is a" relationship.
- Used to represent inheritance in knowledge representation.
- Organizes concepts in hierarchical structure.

Example:
Animal
→ Mammal
→ Dog

Dog ISA Mammal
Mammal ISA Animal

It allows inheritance of properties from parent to child.

Frame Notation

- A frame is a data structure for representing structured knowledge.
- Contains:
    - Frame name
    - Slots (attributes)
    - Slot values

Example Frame:

Frame: Student

- Name: Amit
- Course: BSc
- Age: 20

Frames support inheritance in ISA hierarchy.